Logistics Data Preparation Pipeline: A Comprehensive Guide

1. Data Collection Simulation

For this pipeline, we will simulate the collection of a public logistics dataset, such as a Global Supply Chain Shipping dataset. The dataset includes historical shipments with the following key attributes (columns):

- `order_id`: Unique identifier for the shipment.
- `shipping_date` and `delivery_date`: Timestamps for when the item was shipped and received.
- `freight_cost`: The monetary cost of transporting the goods.
- `item_weight`: The physical weight of the package.
- `destination_port`: The delivery endpoint.

Data Characteristics: The raw data extracted from the Enterprise Resource Planning (ERP) system is messy. It contains null values where sensors failed to record weight, missing delivery dates for items still in transit, and extreme outliers in freight costs due to manual entry errors.

2. Data Cleaning & Preprocessing Methodology

To ensure high-quality analysis and accurate predictive modeling later in the process, several preprocessing techniques must be applied:

- **Handling Missing Values:** Missing data can skew models or cause algorithms to fail.
    - *Method:* If a `delivery_date` is missing, it implies the package hasn't arrived; these records will be filtered out for lead-time analysis. For numerical values like `freight_cost`, missing entries will be imputed using the median (which is robust to outliers) rather than dropping valuable rows.
- **Outlier Detection:** Logistics data is prone to entry errors (e.g., entering 1000 kg instead of 10.00 kg).

    - *Method:* We will use the Interquartile Range (IQR) method on `item_weight`. Any data points falling outside 1.5 times the IQR above the third quartile or below the first quartile will be capped or removed to prevent them from disproportionately influencing model weights.
- **Normalization (Scaling):** Features like `freight_cost` (ranging in thousands of dollars) and `item_weight` (ranging in single digits) operate on different scales.
    - *Method:* Min-Max scaling will transform numerical features to a standard range (0 to 1), ensuring distance-based algorithms (like clustering for delivery zones) treat all features equally.

3. Python Code Documentation

Below is the Python pipeline using `pandas` and `scikit-learn` to execute the preprocessing methodologies outlined above.import pandas as pd

import numpy as np

from sklearn.preprocessing import MinMaxScaler

1. Data Collection Simulation (Loading Raw Data)

Assume the CSV contains columns: ['order_id', 'shipping_date', 'delivery_date', 'freight_cost', 'item_weight']

df = pd.read_csv("logistics_shipping_data.csv")

print("Initial Dataset Size:", df.shape)

2. Handling Missing Values

Drop records where the delivery hasn't happened yet (missing delivery_date)

df = df.dropna(subset=['delivery_date'])

Impute missing 'freight_cost' with the median cost

```
median_cost = df['freight_cost'].median()

df['freight_cost'] = df['freight_cost'].fillna(median_cost)

# 3. Outlier Detection & Removal (IQR Method on 'item_weight')

Q1 = df['item_weight'].quantile(0.25)

Q3 = df['item_weight'].quantile(0.75)

IQR = Q3 - Q1

lower_bound = Q1 - 1.5 * IQR

upper_bound = Q3 + 1.5 * IQR

# Filter out the outliers

df = df[(df['item_weight'] >= lower_bound) & (df['item_weight'] <=
upper_bound)]

print("Dataset Size after Outlier Removal:", df.shape)

# 4. Normalization / Scaling

# Scale numerical columns to a standard range of (0, 1)

scaler = MinMaxScaler()

numerical_cols = ['freight_cost', 'item_weight']

df[numerical_cols] = scaler.fit_transform(df[numerical_cols])

# Display the cleaned and prepared data head

print(df.head())
```

4. Reflection: The Impact of Data Quality

In logistics, analytics drive million-dollar decisions—from fleet sizing to route planning and dynamic pricing. The "garbage in, garbage out" principle heavily applies here.

If missing values are improperly handled, models might underestimate average delivery times, leading to overpromising to customers. If outliers are left unchecked, a single erroneous 10,000 kg package weight could drastically skew capacity planning algorithms, resulting in unoptimized truck loads or unnecessary extra fleet dispatch. Rigorous data preprocessing acts as the foundation of trustworthy supply chain analytics, ensuring that any subsequent AI or statistical models operate on reality rather than noise.